AN OBJECT IS R-PERFECT EXACTLY WHEN IT IS A FINITE COMPLEX OF FLAT FINITELY PRESENTED MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- An object of finite tor amplitude over the base is R-perfect as soon as it is pseudo-coherent. used at proof lines 21990-21995. Reduces the implication from (2) to (1) to showing that a flat finitely presented module is pseudo-coherent.
- A flat finitely presented module over a base descends to a finitely generated situation, where finite type modules over Noetherian rings are pseudo-coherent. invoked at proof lines 21995-22010. Produces the finitely generated model M_0 over a finitely generated base and its pseudo-coherence, and the finite free resolution that spreads out.
- A bounded above complex of finite free modules whose truncation cuts at the tor amplitude lower bound consists of flat finitely presented modules. invoked at proof lines 22015-22019. Supplies the representing complex for the implication from (1) to (2).

Source reference. Original lemma 0DHT: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21949–21949. The sentence at proof lines 21990-21992 invokes it to reduce (2) implies (1) to a single R-flat finitely presented module; the verification of that case is carried out inside the counted span.

Source reference. Original lemma 066E: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16517–16517. Used at proof line 22003 to conclude that the descended module M_0 is pseudo-coherent; the spreading out of M_0, the finite free resolution and the base change back to A are inside the counted span.

Source reference. Original lemma 0661: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14955–14955. Cited parenthetically at proof lines 22008-22009 to see that the base changed finite free resolution still resolves M; the flatness of A_0 over R_0 and the flatness of M_0 over R_0 that make that comparison available are used in the counted span.

Source reference. Original lemma 0653: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16883–16883. Used at proof line 22018 for the implication from (1) to (2): it says the truncated finite free resolution consists of flat finitely presented modules; the truncation step itself and the identification of the representing complex are inside the counted span.

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Source reference. Original lemma 02JO: sources/stacks-chow/tags.txt, source lines 1–1. The reference at proof line 21995 produces the finitely generated model R_0 to A_0 and the module M_0; everything done with that model afterwards, including the resolution and the base change, is inside the counted span.

Lemma 0.2. *Let $R \to A$ be a flat ring map of finite presentation. Let $K \in D(A)$. The following are equivalent*

(1) *K is R-perfect, and*
(2) *K is isomorphic to a finite complex of R-flat, finitely presented A-modules.*

Proof. To prove (2) implies (1) it suffices by Lemma 0DHT to show that an R-flat, finitely presented A-module M defines an R-perfect object of $D(A)$. Since M has finite tor dimension over R, it suffices to show that M is pseudo-coherent. By Algebra, Lemma 02JO there exists a finite type $\mathbf{Z}$-algebra $R_0 \subset R$ and a flat finite type ring map $R_0 \to A_0$ and a finite A_0-module M_0 flat over R_0 such that $A = A_0 \otimes_{R_0} R$ and $M = M_0 \otimes_{R_0} R$. By Lemma 066E we see that M_0 is pseudo-coherent A_0-module. Choose a resolution $P_0^\bullet \to M_0$ by finite free A_0-modules P_0^n. Since A_0 is flat over R_0, this is a flat resolution. Since M_0 is flat over R_0 we find that $P^\bullet = P_0^\bullet \otimes_{R_0} R$ still resolves $M = M_0 \otimes_{R_0} R$. (You can use Lemma 0661 to see this.) Hence $P^\bullet$ is a finite free resolution of M over A and we conclude that M is pseudo-coherent.

Assume (1). We can represent K by a bounded above complex $P^\bullet$ of finite free A-modules. Assume that K viewed as an object of $D(R)$ has tor amplitude in $[a, b]$. By Lemma 0653 we see that $\tau_{\geq a} P^\bullet$ is a complex of R-flat, finitely presented A-modules representing K. □